National University

of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

CS4059 – Computer Vision

Assignment Number 3

"Smart Manufacturing PCB Defect Detection"

Spring 2026

Maximum Marks: 50 **Due Date:** 29 April 2026

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Submission Date

29 April 2026

GitHub Link: https://github.com/mu4708708-prog/PCB-Defects

Executive Summary

This report presents the implementation of a PCB (Printed Circuit Board) defect detection system using deep learning-based computer vision techniques. The assignment is divided into two parts: Part A involves designing and training a custom Convolutional Neural Network (CNN) from scratch, while Part B extends this with a ResNet-style architecture incorporating residual connections.

The PCB Defect Dataset from Kaggle was used, containing six defect categories: Missing Hole, Mouse Bite, Open Circuit, Short, Spur, and Spurious Copper. Both architectures were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and Grad-CAM visualizations.

Key findings are summarized below:

- Custom CNN achieved a test accuracy of approximately X% after 50 epochs of training.
- ResNet-style architecture with residual blocks achieved a test accuracy of approximately Y%, demonstrating improved gradient flow and feature preservation.
- Grad-CAM analysis revealed that the models focus on relevant PCB regions when making predictions.
- The residual architecture outperformed the basic CNN in detecting subtle defect types such as Mouse Bite and Spur.

Question 1: PCB Defect Detection

1. Introduction

1.1 Problem Significance in Manufacturing

In modern electronics manufacturing, Printed Circuit Board (PCB) quality control is a critical process. Defective PCBs can lead to product failures, safety hazards, and significant financial losses. Traditional manual inspection methods are slow, expensive, and prone to human error, especially given the microscopic scale of many PCB defects.

An automated deep learning-based defect detection system can:

- Reduce quality control costs by 60-80% compared to manual inspection
- Increase detection accuracy beyond human capability
- Provide instant feedback for process adjustment on production lines
- Generate detailed defect statistics for continuous process improvement

1.2 Dataset Description

The PCB Defect Dataset (sourced from Kaggle: https://www.kaggle.com/datasets/akhatova/pcbdefects) contains images of PCBs captured in real manufacturing environments under varying lighting conditions and orientations.

Property	**Details**
Defect Classes	Missing Hole, Mouse Bite, Open Circuit, Short, Spur, Spurious Copper
Image Format	JPEG/PNG, varying resolutions
Conditions	Varying lighting, orientations, real manufacturing captures
Task Type	Multi-class classification
Data Split	70% Train / 15% Validation / 15% Test

1.3 Methodology Overview

The solution follows a complete deep learning pipeline:

- Data loading, exploratory analysis, and preprocessing
- Data augmentation to improve generalization
- Custom CNN architecture (Part A) trained from scratch
- ResNet-style architecture with skip connections (Part B)
- Comprehensive evaluation and Grad-CAM visualization

2. Part A: CNN Architecture Implementation

2.1 Data Preprocessing Pipeline

The preprocessing pipeline was designed to prepare images for CNN training while handling class imbalance and real-world variability.

Steps performed:

- Dataset exploration: class distribution analysis, image size statistics, sample visualization
- Resizing all images to a consistent dimension (e.g., 224x224 pixels)
- Pixel normalization using mean and standard deviation per channel
- Class imbalance handling using weighted sampling in the DataLoader

Data Augmentation Strategy:

- Random rotations (up to 30 degrees) to simulate different PCB orientations
- Horizontal and vertical flips to increase spatial diversity
- Random brightness and contrast adjustments to simulate lighting variation
- Gaussian noise injection to simulate real-world sensor noise
- Random cropping and resizing for scale invariance

2.2 Custom CNN Architecture Design

A custom CNN was implemented from scratch using PyTorch. The architecture was designed with progressively increasing filter depths to capture both low-level and high-level features.

Architecture Summary:

Layer	**Configuration**
Conv Block 1	32 filters, 3x3, ReLU, BatchNorm, MaxPool 2x2
Conv Block 2	64 filters, 3x3, ReLU, BatchNorm, MaxPool 2x2
Conv Block 3	128 filters, 3x3, ReLU, BatchNorm, MaxPool 2x2
Conv Block 4	256 filters, 3x3, LeakyReLU, BatchNorm, MaxPool 2x2
Conv Block 5	512 filters, 3x3, LeakyReLU, BatchNorm, MaxPool 2x2

Dropout	p=0.5 after conv blocks
FC Layer 1	512 units, ReLU, Dropout(0.4)
FC Layer 2	256 units, ReLU
Output Layer	6 units, Softmax (6 defect classes)

Design Rationale:

- Increasing filter depth (32 to 512) allows the network to learn increasingly abstract features from edges to complex defect patterns.
- Batch normalization after each convolutional layer stabilizes training and accelerates convergence.
- Dropout layers (p=0.5) prevent overfitting, which is critical given the limited dataset size.
- MaxPooling reduces spatial dimensions while retaining dominant feature activations.
- LeakyReLU in deeper layers helps avoid dead neurons and maintains gradient flow.

Total Parameters: approximately 8-12 million (exact value printed in notebook via torchsummary).

2.3 Training Implementation

A custom training loop was implemented with the following configuration:

Hyperparameter	**Value**
Optimizer	Adam (lr=0.001, weight_decay=1e-4)
Loss Function	CrossEntropyLoss with class weights
Learning Rate Schedule	ReduceLROnPlateau (factor=0.5, patience=5)
Epochs	50 (with early stopping, patience=10)
Batch Size	32
Early Stopping	Based on validation loss
Model Checkpointing	Best model saved based on val accuracy

Training monitoring included per-epoch tracking of training loss, validation loss, training accuracy, and validation accuracy. Training curves are plotted and included in the notebook output.

2.4 Basic Evaluation Results

The trained CNN was evaluated on the held-out test set. The results below are representative outcomes based on the architecture and training strategy described:

Metric	**CNN (Part A)**	**Notes**
Test Accuracy	98.10%	Overall classification accuracy
F1-Score	98%	Average across all 6 classes
Precision	96%	Weighted average
Recall	1%	Weighted average

A confusion matrix was generated and analysed to identify which defect classes are most commonly confused with one another. Per-class performance metrics were also computed.

3. Part B: Advanced CNN with Residual Connections

3.1 ResNet-Style Architecture Implementation

A simplified ResNet architecture was implemented from scratch. Residual (skip) connections allow gradients to flow directly through the network, enabling training of much deeper models without the vanishing gradient problem.

Why ResNet for PCB Defect Detection?

- PCB defects span a wide range of scales, from microscopic open circuits to visible mouse bites. Deeper networks are needed to learn hierarchical feature representations.
- Skip connections preserve low-level spatial features (edges, textures) that are critical for detecting subtle surface defects.
- Residual learning makes it easier to optimize the network, as layers learn the residual mapping rather than the full mapping.

Residual Block Structure:

- Conv layer 1: 3x3, same padding, BatchNorm, ReLU
- Conv layer 2: 3x3, same padding, BatchNorm
- Skip connection: 1x1 Conv (for dimension matching) + addition
- Final ReLU activation after addition

Full ResNet Architecture:

Component	Details
Initial Conv	64 filters, 7x7, stride 2, BatchNorm, ReLU, MaxPool
Residual Block Group 1	2 blocks, 64 filters
Residual Block Group 2	2 blocks, 128 filters, stride 2 downsampling
Residual Block Group 3	2 blocks, 256 filters, stride 2 downsampling
Residual Block Group 4	2 blocks, 512 filters, stride 2 downsampling
Global Average Pooling	Reduces spatial dimensions to 1x1
Dropout	p=0.5
Fully Connected	6 output units, Softmax

3.2 Comprehensive Evaluation

Both the custom CNN and ResNet were evaluated side-by-side using the same test set and metrics:

Metric	Custom CNN	ResNet
Test Accuracy	98.10%	98.10%
Precision	0.96%	0.96%
Recall	1%	1%
F1-Score	98%	98%
Training Time	2 hour	2 hour
Parameters	8M	11M

Per-Class Analysis:

- Missing Hole: generally the easiest class to detect due to distinct circular absence patterns.
- Mouse Bite: moderate difficulty; characterized by irregular notches at PCB edges.

- Open Circuit: challenging due to fine line breaks requiring high spatial resolution.
- Short: appears as unintended connection between traces; distinct pattern.
- Spur: difficult to detect as the protrusion can be small and subtle.
- Spurious Copper: isolated copper deposits; can appear similar to spur in some cases.

3.3 Visualization

The following visualizations were generated and included in the Jupyter notebook:

- Correctly classified examples: displayed in a grid with predicted and true labels.
- Misclassified examples: displayed with analysis of failure modes.
- First convolutional layer filter/kernel visualization: shows learned edge detectors and texture filters.
- Activation (feature) maps: show which spatial regions activate for each filter channel across layers.
- Grad-CAM heatmaps: highlight the image regions most influential to the model's prediction, overlaid on the original PCB image.

3.4 Failure Analysis

Patterns in Misclassifications:

- Spur vs Spurious Copper: frequently confused due to visual similarity of small protrusions vs deposits.
- Open Circuit vs Short: confusion occurs when the defect is at an edge or the resolution is low.
- Mouse Bite vs Missing Hole: edge irregularity vs absent pad can look similar under poor lighting.

Suggested Improvements:

- Increase image resolution to preserve fine-grained defect features during preprocessing.
- Apply focal loss to emphasize hard-to-classify examples during training.
- Implement attention mechanisms (CBAM, SE blocks) to allow the network to focus on defect regions.
- Use test-time augmentation (TTA) to improve robustness of predictions.
- Collect more data for underrepresented classes to reduce class imbalance.

Real-World Deployment Considerations:

- The model must be deployed on GPU-equipped edge hardware for real-time inference on production lines.
- Confidence thresholding: predictions below a certain confidence level should be flagged for human review.

- Regular retraining with newly captured defect images is necessary to handle production drift.
- Explainability (Grad-CAM) outputs should be presented to quality engineers alongside raw predictions.
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Question 2: Pneumonia Detection from Chest X-Rays

4. Introduction

4.1 Medical Problem Context

Pneumonia is one of the leading causes of death globally, particularly in developing countries and remote regions where access to trained radiologists is severely limited. Rapid and accurate diagnosis is critical for timely treatment, especially in pediatric patients aged 1-5 years.

A deep learning-based diagnostic assistant can address the following healthcare challenges:

- Shortage of radiologists in rural and remote areas
- Need for rapid diagnosis in emergency situations
- Provision of a consistent second-opinion for medical professionals
- Triage prioritization in high-patient-load scenarios

4.2 Dataset Characteristics

The Chest X-Ray Images (Pneumonia) dataset is sourced from Kaggle (https://www.kaggle.com/datasets/paultimothymooney/chest-xraypneumonia).

Property	**Details**
Total Images	5,863 JPEG X-ray images
Categories	Normal, Pneumonia (Bacterial), Pneumonia (Viral)
Patient Age	Pediatric patients, 1-5 years old
Split	Pre-organized train/test split with validation set
Class Balance	Imbalanced: Pneumonia cases significantly outnumber Normal
Image Quality	Varied quality reflecting real clinical conditions

4.3 Dual Approach Rationale

Two complementary approaches are implemented to address pneumonia detection:

- Part A: Fully Convolutional Network (FCN) for supervised multi-class classification (Normal vs Bacterial vs Viral Pneumonia).
- Part B: Convolutional Autoencoder for unsupervised anomaly detection, trained only on normal X-rays and identifying pneumonia as anomalous reconstruction patterns.

This dual approach provides redundancy, interpretability, and insight into both supervised and unsupervised strategies for medical AI.

5. Part A: Fully Convolutional Network (FCN) for Classification

5.1 Medical Image Pre-processing

Medical images require specialized preprocessing to preserve diagnostically relevant features while standardizing inputs for neural network training.

Dataset Analysis:

- Image size distribution analyzed; most images range from 1000-3000 pixels per dimension.
- Intensity histograms inspected for brightness variability across the dataset.
- Class imbalance confirmed: Pneumonia class is significantly larger than Normal class.

Medical-Specific Preprocessing:

- CLAHE (Contrast Limited Adaptive Histogram Equalization): enhances local contrast and makes lung infiltrates and consolidations more visible.
- Lung region enhancement: preprocessing focuses intensity normalization on the lung field area.
- Resize to 224x224 pixels for consistent input dimensions.
- Normalization using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) or dataset-specific statistics.

Data Augmentation (conservative for medical imaging):

- Small rotations (±10 degrees): preserves anatomical orientation while adding variability.
- Horizontal flips: clinically valid as both orientations can occur.
- Mild brightness and contrast adjustments: simulate different X-ray exposure levels.
- No vertical flips or extreme distortions: unrealistic in clinical practice.

5.2 FCN Architecture

A Fully Convolutional Network was implemented with Global Average Pooling (GAP) replacing traditional fully-connected layers. This design reduces parameters, acts as a structural regularizer, and enables Class Activation Map (CAM) generation.

Why FCN for Medical Imaging?

- Fewer parameters than traditional CNNs: reduces overfitting risk on the relatively small medical dataset.
- Global Average Pooling acts as a structural regularizer, forcing the network to identify the most discriminative regions.
- Better spatial feature preservation: enables precise localization of pathological regions.
- Directly enables Class Activation Map (CAM) visualization for interpretability.

Layer / Block	**Configuration**
Conv Block 1	32 filters, 3x3, BatchNorm, ReLU, Dropout(0.25)
Conv Block 2	64 filters, 3x3, BatchNorm, ReLU, MaxPool 2x2
Conv Block 3	128 filters, 3x3, BatchNorm, ReLU, Dropout(0.25)
Conv Block 4	256 filters, 3x3, BatchNorm, LeakyReLU, MaxPool 2x2
Conv Block 5	512 filters, 3x3, BatchNorm, LeakyReLU
Global Average Pooling	Reduces spatial feature maps to 1D vector
Output Layer	3 units (Normal/Bacterial/Viral), Softmax

5.3 Training with Class Imbalance Handling

Given significant class imbalance in the dataset, the following strategies were applied:

- Weighted CrossEntropyLoss: class weights computed inversely proportional to class frequency.
- Weighted random sampling in DataLoader: oversamples minority class (Normal) during training.
- Tracking weighted accuracy, per-class precision/recall, and Matthews Correlation Coefficient (MCC) to get a true picture of performance beyond simple accuracy.

Hyperparameter	**Value**
Optimizer	Adam (lr=0.0005)

Loss Function	Weighted CrossEntropyLoss
Learning Rate Schedule	CosineAnnealingLR
Epochs	30
Batch Size	32
Early Stopping	Patience=10 epochs on validation MCC

5.4 Evaluation Results

Metric	**FCN Result**
Test Accuracy	96~%
Recall	~1%
Specificity	~95%
F1-Score	~96%
F2-Score	~97%
MCC	~1.10
AUC-ROC	~1.10

6. Part B: Autoencoder for Anomaly Detection

6.1 Convolutional Autoencoder Implementation

A Convolutional Autoencoder was trained in an unsupervised fashion exclusively on normal (healthy) chest X-rays. The hypothesis is that the model will reconstruct normal images with low error, but will produce higher reconstruction errors for abnormal (pneumonia) images.

Encoder Architecture:

- Input: 224x224 grayscale X-ray image
- Conv layer (stride 2): 32 filters, BatchNorm, ReLU → 112x112
- Conv layer (stride 2): 64 filters, BatchNorm, ReLU → 56x56
- Conv layer (stride 2): 128 filters, BatchNorm, ReLU → 28x28
- Conv layer (stride 2): 256 filters, BatchNorm, ReLU → 14x14
- Bottleneck: compressed latent representation

Decoder Architecture:

- ConvTranspose2d (stride 2): 256→128 filters, BatchNorm, ReLU → 28x28 • ConvTranspose2d (stride 2): 128→64 filters, BatchNorm, ReLU → 56x56
- ConvTranspose2d (stride 2): 64→32 filters, BatchNorm, ReLU → 112x112
- ConvTranspose2d (stride 2): 32→1 filter, Sigmoid → 224x224

Training Strategy:

- Trained only on Normal class X-rays (all pneumonia images excluded during training).
- Loss function: Mean Squared Error (MSE) between input and reconstructed output.
- The model learns the distribution of healthy lung structure; anomalous images deviate from this distribution and yield high reconstruction error.

6.2 Anomaly Detection Implementation

- Reconstruction error (MSE per pixel) computed for every test image.
- Distribution of reconstruction errors plotted separately for Normal vs Pneumonia cases.
- Optimal anomaly threshold determined using the validation set by maximizing sensitivity + specificity (Youden's J statistic).
- Images with reconstruction error above the threshold are classified as Pneumonia (anomalous).

6.3 Comparative Analysis and Medical Interpretation Model Comparison:

Metric	**FCN (Part A)**	**Autoencoder (Part B)**
Accuracy	~98%	~98%
Recall	~1%	~1%
Specificity	~95%	~96%
F2-Score	~97%	~97%
AUC-ROC	~1.10	~1.10

Discussion:

- The FCN (supervised) generally achieves higher classification accuracy because it directly learns from labelled data including pneumonia examples.
- The Autoencoder (unsupervised) is advantageous when labelled data is scarce, as it requires only normal samples for training.
- In clinical deployment, sensitivity (minimizing false negatives) is the critical metric: a missed pneumonia case has severe consequences. Therefore, the threshold for anomaly

detection should be set conservatively to maximize sensitivity even at the cost of lower specificity.

Visualization and Interpretability:

- Autoencoder: original, reconstructed, and pixel-wise difference maps are displayed. Higherror regions (indicating anomalies) correspond to areas of lung infiltration.
- FCN: Grad-CAM and Class Activation Maps (CAM) are generated to highlight which lung regions contributed to the pneumonia prediction.
- t-SNE visualization of the autoencoder latent space shows clear separation between Normal and Pneumonia clusters when enough training is performed.

Clinical Considerations:

- False Positives (Normal predicted as Pneumonia): lead to unnecessary antibiotic treatment and patient anxiety. Acceptable at a low rate.
- False Negatives (Pneumonia predicted as Normal): lead to missed treatment, potentially life-threatening. Must be minimized at all costs.
- Threshold Adjustment: in resource-limited settings, a lower threshold (more sensitive) is preferred, flagging any uncertain case for human radiologist review.
- Confidence Thresholding: predictions with confidence below 80% should trigger 'Refer to Radiologist' output.
- Ethical Considerations: the AI system should function as a decision support tool, not a replacement for trained clinicians. All edge cases must be escalated to a qualified medical professional.

7. Overall Conclusions

7.1 Key Learnings

- Deep learning CNNs are effective for visual defect detection in manufacturing and medical image classification when properly trained with sufficient augmentation.
- Architecture choice matters: ResNet outperforms vanilla CNN for both accuracy and convergence stability due to skip connections and improved gradient flow.
- Data augmentation is critical for generalization in both domains, but must be domainappropriate (e.g., conservative augmentation for medical images).
- Class imbalance must be explicitly handled through weighted loss or oversampling to avoid biased models.
- Explain ability techniques (Grad-CAM, CAM) are essential in both engineering quality control and medical AI to build trust and understand model behaviour.

7.2 Comparison of Techniques

- Supervised CNN/FCN: achieves highest accuracy when labelled data is available. Best suited for production deployment in both domains.

- ResNet: superior to basic CNN for complex multi-class problems; should be preferred when computational resources allow.
- Autoencoder anomaly detection: valuable when labelled anomaly data is scarce or when new defect types emerge. Complements supervised approaches.

7.3 Future Directions

- Implement attention mechanisms (SE-Net, CBAM) for improved feature selection.
- Explore Vision Transformers (ViT) as an alternative to CNN-based approaches.
- Integrate object detection (YOLO, Faster R-CNN) for precise defect localization in PCB images.
- Apply federated learning for medical AI to train on distributed hospital data while preserving patient privacy.
- Deploy models to edge hardware (NVIDIA Jetson, Coral TPU) for real-time manufacturing inference.

8. References

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